

Name: _____

Class: _____

Date: _____

Homework 1.1

Collaboration

Score: ____ / 8

Multiple Choice

1. Why are computing innovations often developed by teams?

- A. Teams always finish faster than individuals
- B. Only teams are allowed to write code
- C. People with different skills and perspectives can produce better solutions
- D. Teams never make mistakes

2. Which is an example of a computing innovation?

- A. A paper map
- B. A printed calendar
- C. A wooden chair
- D. A GPS navigation app

3. A team designing a school app includes students from every grade. What is the main benefit?

- A. The app will have fewer lines of code
- B. More perspectives help the team identify the needs of more users
- C. The app will not need testing
- D. Only seniors will use the app

4. In pair programming, one person types (the driver) while the other reviews (the navigator). What is a key benefit?

- A. Only one person needs to understand the code
- B. No documentation is needed
- C. The program runs faster
- D. Errors are more likely to be caught as the code is written

Written 1

Describe two specific ways that collaboration could improve an app that helps students find tutoring help.

Written 2

Explain the two roles in pair programming and one benefit of each role.

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Homework 1.2

Program Function and Purpose

Score: _____ / 8

Multiple Choice

1. A music app lets users build playlists of their favorite songs. Which best describes its **purpose**?

- A. It stores song files
- B. It is written in Python
- C. It has a play button
- D. It helps people organize and enjoy the music they like

2. Which is an **output**?

- A. A user typing a search term
- B. A tap on the screen
- C. A notification sound
- D. A GPS reading

3. Which is an **event**?

- A. The program's source code
- B. A user clicking a button
- C. A variable named score
- D. The computer's memory

4. A program reads data from a light sensor every second. The light sensor reading is...

- A. an output
- B. an input
- C. a code segment
- D. a comment

Written 1

Describe the program in a vending machine. Identify two inputs, two outputs, and one event.

Written 2

Write the purpose and one function of a school lunch-ordering app. Then explain how they are different.

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Homework 1.3

Program Design and Development

Score: _____ / 8

Multiple Choice

1. Iterative development means...

- A. writing the whole program at once
- B. refining and revising based on feedback, testing, or reflection
- C. never changing code once written
- D. coding without a plan

2. Which activity is part of the **investigation** phase?

- A. Surveying users about what they need
- B. Fixing a syntax error
- C. Choosing a font for comments
- D. Compressing a file

3. What is the main purpose of comments in code?

- A. To explain the code to people; comments do not affect how the program runs
- B. To make the program run faster
- C. To create outputs
- D. To store data

4. Incremental development means...

- A. building and testing small pieces before adding them to the whole
- B. rewriting everything after each test
- C. skipping the design phase
- D. only working on the user interface

Written 1

Describe how you would develop a "homework tracker" app using BOTH incremental and iterative development.

Written 2

Write a helpful comment for the line `avg ← total / count`. Then explain when a comment must also give credit.

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Homework 1.4

Identifying and Correcting Errors

Score: _____ / 8

Multiple Choice

1. A program will not run because of a missing parenthesis. What type of error is this?

- A. Syntax error
- B. Logic error
- C. Run-time error
- D. Overflow error

2. A program runs but displays the wrong total. What type of error is this?

- A. Syntax error
- B. Logic error
- C. Overflow error
- D. No error

3. A program crashes while running when it tries to divide by zero. What type of error is this?

- A. Syntax error
- B. Roundoff error
- C. Logic error
- D. Run-time error

4. What is **hand tracing**?

- A. Typing code by hand
- B. Tracking the values of variables step by step on paper
- C. Drawing pictures of the computer
- D. Signing your code

Written 1

This code should count passing grades (70 or higher):

```
count ← 0
FOR EACH g IN grades
{
  IF (g > 70)
  { count ← count + 1 }
}
```

Identify the type of error, give a test list that reveals it, and fix it.

Written 2

Explain the difference between a run-time error and a logic error. Give an example of each.
